

Improving NaNDA Discoverability: Removing Barriers from a Research Gem

How can metadata descriptions support data management, data quality, data discovery, and data use? How might we construct metadata improvements to better connect Social Science researchers to a research gem – The National Neighborhood Data Archive (NaNDA)?

The Problem

- NaNDA has grown rapidly in both scope and visibility, **outpacing the infrastructure** needed to ensure that datasets are easily discoverable, understandable, and usable
- Metadata often determines the **findability, accessibility, interoperability, and reusability (FAIR)** of research data management overall.
- Underutilized datasets can lead to **lost research potential**, wasting public and grant funding, and barriers to data intended for broad public use

The Methodology

User Research

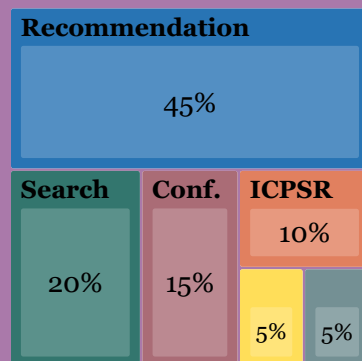
- Four Interviews with NaNDA Users
- Search Behavior Survey (20 responses)
- Exploratory User Metric Analysis

MAP

- Search Terms Crosswalk Data Creator
- Guide for researchers on how to explain data

The Findings

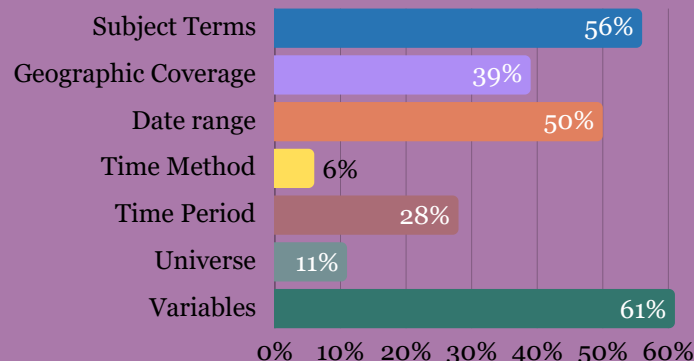
How Do Researchers First Encounter NaNDA?



User Voices - What Did Researchers Say?

- "Discovering new data is difficult - especially data housed in openICPSR."
- "It's intuitive once you know what to look for, but not for a first-time user without connections."
- "Finding the dataset was easy, but identifying the variables is more difficult."

When searching the NaNDA repository, which fields or filters do Researchers use most often?



Metadata Application Profile for Specialized Research Communities

User Search Term	Research Domain	ICPSR Subject Terms	JEL Codes	MeSH Terms	Notes / Neighborhood Context
Police & Law Enforcement	Criminology / Public Policy	police police-community relations police misconduct assaults on police arrest rates criminal justice policy	K42 - Illegal Behavior and Enforcement D73 - Bureaucracy, Administrative Processes H56 - National Security	Police Law Enforcement Enforcement Criminal Justice Community-Institutional Relations Police-Community Relations	Officer deployment, use-of-force incidents, stop-and-frisk data, community trust surveys
Immigration & Migration	Demography / Economics	immigrants immigration status immigration policy refugees foreign born migration acculturation ethnicity	F22 - International Migration J15 - Economics of Minorities, Immigrants J61 - Geographic Labor Mobility, Immigrant Workers R23 - Regional Migration	Emigrants and Immigrants Transients and Migrants Refugees Acculturation Immigration Policy Foreign-Born Population	Foreign-born share, country of origin, naturalization rates, immigrant enclave formation
Gentrification & Displacement	Urban Planning / Sociology	gentrification neighborhood change housing costs displacement neighborhood conditions residential mobility	R11 - Regional Economic Activity, Growth, Development R31 - Housing Supply and Markets J61 - Geographic Labor Mobility D31 - Personal Income, Wealth	Gentrification Residence Characteristics Urban Renewal Markets Neighborhood Change	Rent appreciation, demographic shifts, business turnover, long-term resident displacement indicators

Domain Group	Color
Crime & Safety	
Race, Equity & Demo.	
Housing	
Education	
Health	
Environment & Climate	
Economic Development	
Infrastructure & Transit	
Aging & Disability	
Gender & Family	
Social Services	



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The Solutions

Improving communication across NaNDA's key stakeholders through three different aspects:

- Creating guidelines through a **Metadata Application Profile (MAP)** for researchers and depositors with detailed explanations and suggestions on how to describe a dataset
- Crosswalks** across popular controlled vocabularies in ICPSR such as **MeSH**, **JEL**, and **ICPSR Subject Terms**
- Conducting research into **user search patterns** through interviews with NaNDA users and researchers and an anonymous survey

Acknowledgements

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